Basic Java Concepts

- **What is Java?** Java is a high-level, object-oriented, and platform-independent programming language.
- **What is the JVM?** JVM stands for Java Virtual Machine, which executes compiled Java bytecode.
- **What is a class?** A class is a blueprint or template used to create objects.
- **What is an object?** An object is a runtime instance of a class containing state and behavior.
- **What is a constructor?** A special method invoked automatically when an object is created to initialize its variables.
- **What is inheritance?** A mechanism where one class inherits properties and methods from another class.
- **What is polymorphism?** The ability of a method or object to take multiple forms.
- **What is an interface?** A reference type that contains only abstract methods and constants to achieve abstraction.
- **What is an exception?** An unwanted runtime event that disrupts the normal execution flow of a program.
- **What is a package?** A container used to group related classes and interfaces together.

Slip 1

- **What is a command-line argument?** Values passed to the main method from the terminal during execution using the `args` array.
- **What is an array of objects?** A collection where each index holds a reference to a separate object of a specific class.

Slip 2

- **What does `compareToIgnoreCase()` do?** It compares two strings lexicographically while ignoring case differences.
- **How do you find the size of an array in Java?** By using the built-in `.length` property.

Slip 4

- **What is a character stream class?** Classes like `FileReader` and `FileWriter` used to handle 16-bit Unicode character data.
- **Why should you close files after operations?** To release system resources and prevent memory leaks.

Slip 5

- **What is an abstract class?** A class declared with `abstract` that cannot be directly instantiated.
- **What is method overriding?** Providing a specific subclass implementation for a method already defined in its superclass.

Slip 7

- **What is a custom exception?** A user-defined error type created by extending the built-in `Exception` class.
- **What is the use of the `throw` keyword?** It is used to explicitly trigger and throw an exception object.

Slip 9

- **What is an interface?** A reference type containing abstract methods and constants used to achieve total abstraction.
- **What does the `implements` keyword do?** It indicates that a class is adopting an interface and defining its abstract methods.

Slip 14

- **What is a package in Java?** A namespace grouping related classes and interfaces together into a structured folder.
- **How do you use a class from another package?** By using the `import` statement at the top of your program.

Slip 15

- **What is an applet?** A small Java program that could run inside a web browser supporting Java.
- **What is the lifecycle of an applet?** It consists of four main methods: `init()`, `start()`, `stop()`, and `destroy()`.

Slip 16

- **What is dynamic method dispatch?** A mechanism where a call to an overridden method is resolved at runtime based on the object type.

- **What is the purpose of constructor overloading?** To allow object creation with different sets of initial parameters.

Slip 17

- **What is an interface method by default?** All methods declared inside an interface are implicitly `public` and `abstract`.
- **Can an interface inherit another interface?** Yes, an interface can extend another interface using the `extends` keyword.

Slip 18

- **What is `LinkedList`?** A linear data structure that stores elements in doubly-linked nodes for fast insertions and deletions.
- **How do you add an element at the beginning of a `LinkedList`?** By using the built-in `addFirst()` method.

Slip 19

- **What is the `final` keyword used for?** To declare constants, prevent method overriding, or restrict class inheritance.
- **Can a `final` variable be reassigned?** No, its value becomes a constant once initialized.

Slip 20

- **What is multithreading?** The concurrent execution of multiple threads to achieve maximum CPU utilization.
- **How do you start a thread in Java?** By calling the `.start()` method, which automatically invokes the `run()` method.

Slip 22

- **What is an `ArrayList`?** A resizable array implementation of the `List` interface providing fast random access.
- **What is the difference between `Iterator` and `ListIterator`?** `Iterator` traverses elements in forward direction only, while `ListIterator` supports bidirectional traversal.

Slip 23

- **What is the role of the `super` keyword?** It is used to access immediate parent class variables, methods, or constructors.
- **Can `super()` be used anywhere in a constructor?** No, it must always be the very first statement inside a subclass constructor.

Slip 24

- **What is polymorphism?** The ability of a variable, function, or object to take on multiple forms.
- **What is the difference between compile-time and runtime polymorphism?** Compile-time uses method overloading, while runtime uses method overriding.